

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Artificial Intelligence and Data Science
ASSESSMENT TOOL 2 – SCENARIO BASED

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO2 – Manipulation	Assessment:	Assessment Tool 2 – SCENARIO BASED
Weightage:	35% of CIA		
CO5 Statement:	Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3)		
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Section / Batch:	2024	Date:	13-08-26

Code of Conduct

I M Poojitha Reddy - 192472352 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: M Poojitha Reddy

Instructions

- Read all questions carefully before attempting the answers.
- Answer all five questions. Each question carries equal marks.
- Show all intermediate steps, calculations, probability computations, tagging decisions, and justifications wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
English Word Classes, Tagsets for English, Stochastic Part of Speech Tagging, HMM	Morphological parsing of disagree, agreement, and agreeable; identification of prefixes, suffixes, base forms, and semantic effects of derivation.	Q1

Annexure A – SCENARIO BASED

A company is developing an NLP-based grammar checker for educational applications. The system is trained using a manually POS-tagged corpus. Your task is to recreate the Hidden Markov Model (HMM) **without using any built-in NLP or HMM libraries**.

From the given corpus,

- determine the **Emission Probability**
- determine the **Transition Probability**
- implement the **Viterbi Algorithm**
- predict the POS tags for the new input sentence.

Use only Python dictionaries, loops, and basic programming constructs.

Training Corpus

Sentence	POS Tags
The/DT boy/NN eats/VBZ rice/NN	DT NN VBZ NN
The/DT girl/NN drinks/VBZ milk/NN	DT NN VBZ NN
A/DT cat/NN drinks/VBZ milk/NN	DT NN VBZ NN
The/DT dog/NN chases/VBZ cat/NN	DT NN VBZ NN
A/DT teacher/NN teaches/VBZ students/NNS	DT NN VBZ NNS
Students/NNS study/VBP English/NN	NNS VBP NN
Birds/NNS fly/VBP high/RB	NNS VBP RB
Children/NNS play/VBP games/NNS	NNS VBP NNS

New Input Sentence

The cat drinks milk

Tasks

1. Separate each sentence into words and POS tags.
2. Compute the **Emission Probability** of every word.
3. Compute the **Transition Probability** between POS tags.
4. Recreate the Hidden Markov Model using Python dictionaries.
5. Implement the **Viterbi Algorithm** without using any built-in HMM/NLP libraries.
6. Predict the POS tag sequence for “The cat drinks milk”

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts

Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Q. No. / Criteria Level	Understanding of Scenario	Identification of Key Issues	Application of Concepts	Decision Making	Clarity of Response	Sub. Total	Total %
1	4	4	4	3	4	19	95
2	4	4	4	3	4	19	
3	4	4	4	3	4	19	
4	4	4	4	3	4	19	
5	4	4	4	3	4	19	

Faculty In-charge	Course Coordinator	Program Director
Dr T Kumaragurubaran		
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

ANALYTICAL PROBLEM SOLVING

QUESTION 1: HMM-Based Part-of-Speech Tagging

Title

Morphological Parsing and Stochastic POS Tagging using Hidden Markov Model

1. Problem Understanding

An NLP-based grammar checker needs to predict the correct Part-of-Speech (POS) tag sequence for a new sentence using a manually tagged training corpus.

A **Hidden Markov Model (HMM)** is used where:

- **States** = POS tags
- **Observations** = Words
- **Emission Probability** = Probability of a word given a POS tag
- **Transition Probability** = Probability of one POS tag following another POS tag
- **Viterbi Algorithm** = Finds the most probable sequence of hidden POS tags for the given sentence.

The new sentence is:

“The cat drinks milk”

The required POS tag sequence is obtained using the probabilities learned from the training corpus.

2. Separation of Training Sentences into Words and POS Tags

Sentence 1

The/DT boy/NN eats/VBZ rice/NN
Words: The, boy, eats, rice
Tags : DT, NN, VBZ, NN

Sentence 2

The/DT girl/NN drinks/VBZ milk/NN
Words: The, girl, drinks, milk
Tags : DT, NN, VBZ, NN

Sentence 3

A/DT cat/NN drinks/VBZ milk/NN
Words: A, cat, drinks, milk
Tags : DT, NN, VBZ, NN

Sentence 4

The/DT dog/NN chases/VBZ cat/NN
Words: The, dog, chases, cat
Tags : DT, NN, VBZ, NN

Sentence 5

A/DT teacher/NN teaches/VBZ students/NNS
Words: A, teacher, teaches, students
Tags : DT, NN, VBZ, NNS

Sentence 6

Students/NNS study/VBP English/NN
Words: Students, study, English
Tags : NNS, VBP, NN

Sentence 7

Birds/NNS fly/VBP high/RB
Words: Birds, fly, high
Tags : NNS, VBP, RB

Sentence 8

Children/NNS play/VBP games/NNS
Words: Children, play, games
Tags : NNS, VBP, NNS

3. POS Tag Frequency

The total frequency of each POS tag is calculated from the corpus.

POS Tag	Meaning	Frequency
DT	Determiner	5

POS Tag	Meaning	Frequency
NN	Noun	10
VBZ	Verb, 3rd person singular	5
NNS	Plural noun	5
VBP	Verb, non-3rd person singular	3
RB	Adverb	1
Total		29

Therefore,

Total tagged words = 29

4. Emission Probability

The emission probability is:

$$P(\text{word}|\text{tag}) = \frac{\text{Count}(\text{word}, \text{tag})}{\text{Count}(\text{tag})}$$

It represents the probability of observing a particular word when the POS tag is known.

4.1 Emission Probabilities for DT

Total DT occurrences = 5.

The

Count(The, DT) = 3

$$P(\text{The}|\text{DT}) = \frac{3}{5} = 0.60$$

A

Count(A, DT) = 2

$$P(A|DT)=\frac{2}{5}=0.40$$

4.2 Emission Probabilities for NN

Total NN occurrences = 10.

Word	Count	Emission Probability
boy	1	$1/10 = 0.10$
girl	1	$1/10 = 0.10$
cat	2	$2/10 = 0.20$
milk	2	$2/10 = 0.20$
dog	1	$1/10 = 0.10$
teacher	1	$1/10 = 0.10$
English	1	$1/10 = 0.10$
rice	1	$1/10 = 0.10$

4.3 Emission Probabilities for VBZ

Total VBZ occurrences = 5.

Word	Count	Emission Probability
eats	1	$1/5 = 0.20$
drinks	2	$2/5 = 0.40$
chases	1	$1/5 = 0.20$
teaches	1	$1/5 = 0.20$

4.4 Emission Probabilities for NNS

Total NNS occurrences = 5.

Word	Count	Emission Probability
students	1	$1/5 = 0.20$
Students	1	$1/5 = 0.20$

Word	Count	Emission Probability
Birds	1	$1/5 = 0.20$
Children	1	$1/5 = 0.20$
games	1	$1/5 = 0.20$

4.5 Emission Probabilities for VBP

Total VBP occurrences = 3.

Word	Count	Emission Probability
study	1	$1/3 = 0.3333$
fly	1	$1/3 = 0.3333$
play	1	$1/3 = 0.3333$

4.6 Emission Probability for RB

Total RB occurrences = 1.

$\text{Count}(\text{high}, \text{RB}) = 1$

[
 $P(\text{high}|\text{RB}) = \frac{1}{1} = 1.0$
]

5. Emission Probabilities Required for the New Sentence

The new sentence is:

The cat drinks milk

The required emission probabilities are:

Word 1: The

The is observed with DT three times.

[
 $P(\text{The}|\text{DT})=\frac{3}{5}=0.60$
]

Word 2: cat

cat is observed as NN two times.

[
 $P(\text{cat}|\text{NN})=\frac{2}{10}=0.20$
]

Word 3: drinks

drinks is observed as VBZ two times.

[
 $P(\text{drinks}|\text{VBZ})=\frac{2}{5}=0.40$
]

Word 4: milk

milk is observed as NN two times.

[
 $P(\text{milk}|\text{NN})=\frac{2}{10}=0.20$
]

Therefore:

Word Most Likely Tag Emission Probability

The	DT	0.60
cat	NN	0.20
drinks	VBZ	0.40
milk	NN	0.20

6. Transition Probability

The transition probability is:

[
 $P(t_i|t_{i-1})=$
 $\frac{\text{Count}(t_{i-1}, t_i)}{\text{Count}(t_{i-1})}$
]

{Count(t_{i-1})|text{ with an outgoing transition}}}

The transitions observed in the corpus are counted below.

6.1 Transition Counts

Previous Tag	Next Tag	Count
DT	NN	5
NN	VBZ	5
VBZ	NN	4
VBZ	NNS	1
NNS	VBP	3
VBP	NN	1
VBP	RB	1
VBP	NNS	1

7. Transition Probability Calculations

DT → NN

All 5 DT tags are followed by NN.

[
 $P(\text{NN}|\text{DT}) = \frac{5}{5} = 1.0$
]

NN → VBZ

All 5 NN tags that have a following word are followed by VBZ.

[
 $P(\text{VBZ}|\text{NN}) = \frac{5}{5} = 1.0$
]

VBZ → NN

There are 5 VBZ tags with outgoing transitions.

4 are followed by NN.

$$\begin{aligned} &[\\ &P(\text{NN}|\text{VBZ})=\frac{4}{5}=0.8 \\ &] \end{aligned}$$

VBZ → NNS

1 VBZ is followed by NNS.

$$\begin{aligned} &[\\ &P(\text{NNS}|\text{VBZ})=\frac{1}{5}=0.2 \\ &] \end{aligned}$$

NNS → VBP

All 3 NNS tags with following words are followed by VBP.

$$\begin{aligned} &[\\ &P(\text{VBP}|\text{NNS})=\frac{3}{3}=1.0 \\ &] \end{aligned}$$

VBP → NN

$$\begin{aligned} &[\\ &P(\text{NN}|\text{VBP})=\frac{1}{3}=0.3333 \\ &] \end{aligned}$$

VBP → RB

$$\begin{aligned} &[\\ &P(\text{RB}|\text{VBP})=\frac{1}{3}=0.3333 \\ &] \end{aligned}$$

VBP → NNS

$$\begin{aligned} &[\\ &P(\text{NNS}|\text{VBP})=\frac{1}{3}=0.3333 \\ &] \end{aligned}$$

8. Transition Probability Table

Previous Tag	DT	NN	VBZ	NNS	VBP	RB
DT	0	1.000	0	0	0	0
NN	0	0	1.000	0	0	0
VBZ	0	0.800	0	0.200	0	0
NNS	0	0	0	0	1.000	0
VBP	0	0.333	0	0.333	0	0.333
RB	0	0	0	0	0	0

9. Initial State Probabilities

The first tag of each training sentence is counted.

The first tags are:

DT, DT, DT, DT, DT, NNS, NNS, NNS

Therefore:

DT = 5
 NNS = 3
 Total sentences = 8

Hence,

[
 $P(\text{DT}) = \frac{5}{8} = 0.625$
]

[
 $P(\text{NNS}) = \frac{3}{8} = 0.375$
]

All other initial probabilities are zero.

10. Hidden Markov Model

The HMM consists of:

States

DT, NN, VBZ, NNS, VBP, RB

Observations

The, A, boy, girl, cat, dog, teacher, ...

Initial Probability

$P(DT) = 0.625$
 $P(NNS) = 0.375$

Transition Probability

DT → NN = 1.000

NN → VBZ = 1.000

VBZ → NN = 0.800

VBZ → NNS = 0.200

NNS → VBP = 1.000

VBP → NN = 0.3333

VBP → RB = 0.3333

VBP → NNS = 0.3333

Emission Probability for the New Sentence

$P(\text{The} | DT) = 0.60$

$P(\text{cat} | NN) = 0.20$

$P(\text{drinks} | VBZ) = 0.40$

$P(\text{milk} | NN) = 0.20$

11. Viterbi Algorithm

The Viterbi algorithm calculates the most probable tag sequence using:

$$\begin{aligned} & [\\ & V_t(s) = \\ & \max_{s'} [V_{t-1}(s') P(s|s')] \times P(\text{word}_t | s) \\ &] \end{aligned}$$

where:

- $(V_t(s))$ = maximum probability of a sequence ending in state (s)
- $(P(s|s'))$ = transition probability
- $(P(\text{word}_t | s))$ = emission probability

12. Viterbi Calculation

Input:

The cat drinks milk

Expected candidate sequence:

DT → NN → VBZ → NN

Step 1: Word = “The”

For DT:

$$\begin{aligned} & [\\ & V_1(DT) = P(DT) \times P(\text{The}|DT) \\ &] \end{aligned}$$

$$\begin{aligned} & [\\ & = 0.625 \times 0.60 \\ &] \end{aligned}$$

$$\begin{aligned} & [\\ & = 0.375 \\ &] \end{aligned}$$

Therefore:

$$V1(DT) = 0.375$$

Other tags have zero emission probability for The.

Step 2: Word = “cat”

For NN:

$$\begin{aligned} & [\\ & V_2(NN) = V_1(DT) \times P(NN|DT) \times P(\text{cat}|NN) \\ &] \end{aligned}$$

$$[\\ =0.375 \times 1.0 \times 0.20 \\]$$

$$[\\ =0.075 \\]$$

Therefore:

$$V_2(NN) = 0.075$$

Step 3: Word = “drinks”

For VBZ:

$$[\\ V_3(VBZ) = V_2(NN) \times P(VBZ|NN) \times P(drinks|VBZ) \\]$$

$$[\\ =0.075 \times 1.0 \times 0.40 \\]$$

$$[\\ =0.030 \\]$$

Therefore:

$$V_3(VBZ) = 0.030$$

Step 4: Word = “milk”

For NN:

$$[\\ V_4(NN) = V_3(VBZ) \times P(NN|VBZ) \times P(milk|NN) \\]$$

$$[\\ =0.030 \times 0.80 \times 0.20 \\]$$

[
=0.0048
]

Therefore:

V4 (NN) = 0.0048

13. Viterbi Table

Position	Word	Best POS Tag	Viterbi Probability
1	The	DT	0.375
2	cat	NN	0.075
3	drinks	VBZ	0.030
4	milk	NN	0.0048

Final probability:

[
P(The,cat,drinks,milk,\DT,NN,VBZ,NN)=0.0048
]

14. Backtracking

The Viterbi backtracking gives:

milk → NN
drinks → VBZ
cat → NN
The → DT

Reversing the sequence:

DT → NN → VBZ → NN

15. Final POS Tag Prediction

Therefore, the predicted POS tags for:

The cat drinks milk

are:

The/DT cat/NN drinks/VBZ milk/NN

Final Answer

```
[  
\boxed{DT\NN\VBZ\NN}  
]
```

16. Python Implementation of HMM and Viterbi Algorithm

The following program uses only **Python dictionaries, loops, and basic programming constructs**. No NLP or HMM library is used.

```
corpus = [  
    [("The", "DT"), ("boy", "NN"), ("eats", "VBZ"), ("rice", "NN")],  
    [("The", "DT"), ("girl", "NN"), ("drinks", "VBZ"), ("milk", "NN")],  
    [("A", "DT"), ("cat", "NN"), ("drinks", "VBZ"), ("milk", "NN")],  
    [("The", "DT"), ("dog", "NN"), ("chases", "VBZ"), ("cat", "NN")],  
    [("A", "DT"), ("teacher", "NN"), ("teaches", "VBZ"), ("students", "NNS")],  
    [("Students", "NNS"), ("study", "VBP"), ("English", "NN")],  
    [("Birds", "NNS"), ("fly", "VBP"), ("high", "RB")],  
    [("Children", "NNS"), ("play", "VBP"), ("games", "NNS")]  
]  
  
tags = ["DT", "NN", "VBZ", "NNS", "VBP", "RB"]  
  
tag_count = {}  
word_tag_count = {}  
transition_count = {}  
transition_total = {}  
initial_count = {}  
  
for sentence in corpus:  
    first_tag = sentence[0][1]  
    initial_count[first_tag] = initial_count.get(first_tag, 0) + 1  
  
    for word, tag in sentence:  
        tag_count[tag] = tag_count.get(tag, 0) + 1  
        key = (word, tag)  
        word_tag_count[key] = word_tag_count.get(key, 0) + 1  
  
    for i in range(len(sentence) - 1):  
        t1 = sentence[i][1]
```

```

        t2 = sentence[i + 1][1]

        key = (t1, t2)
        transition_count[key] = transition_count.get(key, 0) + 1
        transition_total[t1] = transition_total.get(t1, 0) + 1

    emission = {}

    for (word, tag), count in word_tag_count.items():
        emission[(word, tag)] = count / tag_count[tag]

    transition = {}

    for (t1, t2), count in transition_count.items():
        transition[(t1, t2)] = count / transition_total[t1]

    initial = {}

    for tag in tags:
        initial[tag] = initial_count.get(tag, 0) / len(corpus)

    sentence = ["The", "cat", "drinks", "milk"]

    viterbi = [{}]
    backpointer = [{}]

    for tag in tags:
        viterbi[0][tag] = initial[tag] * emission.get((sentence[0], tag), 0)
        backpointer[0][tag] = None

    for i in range(1, len(sentence)):
        viterbi.append({})
        backpointer.append({})

        for current_tag in tags:
            best_probability = 0
            best_previous = None

            emit = emission.get((sentence[i], current_tag), 0)

            for previous_tag in tags:
                trans = transition.get((previous_tag, current_tag), 0)

                probability = (
                    viterbi[i - 1][previous_tag]
                    * trans
                    * emit
                )

                if probability > best_probability:
                    best_probability = probability
                    best_previous = previous_tag

            viterbi[i][current_tag] = best_probability
            backpointer[i][current_tag] = best_previous

    best_tag = None

```

```
best_probability = 0

for tag in tags:
    if viterbi[-1][tag] > best_probability:
        best_probability = viterbi[-1][tag]
        best_tag = tag

result = [best_tag]

for i in range(len(sentence) - 1, 0, -1):
    result.append(backpointer[i][result[-1]])

result.reverse()

print("Sentence:", " ".join(sentence))
print("POS Tags:", result)
print("Probability:", best_probability)
```

17. Program Output

```
Sentence: The cat drinks milk
POS Tags: ['DT', 'NN', 'VBZ', 'NN']
Probability: 0.0048
```

Therefore:

```
The/DT cat/NN drinks/VBZ milk/NN
```

18. Conclusion

The HMM was successfully constructed from the manually POS-tagged corpus by calculating the **initial, emission, and transition probabilities**. The Viterbi algorithm was then applied to the new sentence.

The most probable POS tag sequence is:

```
[
\boxed{DT\->NN->VBZ->NN}
]
```

Hence, the final POS-tagged sentence is:

```
The/DT cat/NN drinks/VBZ milk/NN
```

The result demonstrates that the HMM can learn POS patterns from the training corpus and use them to predict the most probable grammatical structure of a new sentence.